

APPENDIX K
DERIVATION OF EQUATION FOR
EVAPORATION FROM OPEN SURFACES
(REVISED)

Evaporation of a Liquid in a Flowing Air Stream

Evaporation can be defined as the change of state of a liquid into a gas at the cost of a specific amount of energy. Evaporation of a liquid in a flowing air stream can be idealized as in Figure 1, in which the air flow direction is defined to be along the z axis, and the flow is always parallel to the liquid surface.

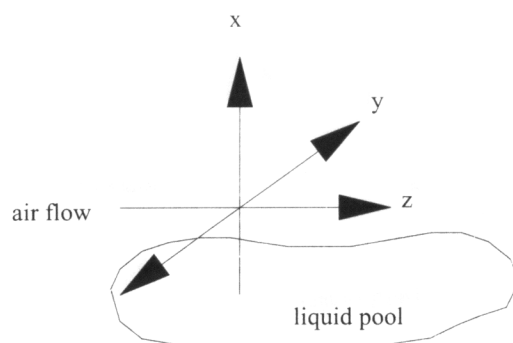


Figure 1

Once the evaporation system reaches equilibrium, namely, the surface temperature of a sufficiently large pool is constant and the heat of evaporation is provided by the surroundings, the evaporation rate will also be a constant. In this case, we can establish a mass balance on a differential element above the liquid pool (normal to the liquid surface, or in the x direction) and along the air flow (z direction). A similar system has been considered in a standard text of Transport Phenomena by Bird Stewart and Lightfoot (Section 17.5).

Along a volume element, the following equation applies:

$$\frac{\delta N_{A,z}}{\delta z} + \frac{\delta N_{A,x}}{\delta x} = 0 \quad (1)$$

where $N_{A,z}$ and $N_{A,x}$ are the molar fluxes of chemical A in the z or x direction (moles/unit time/unit area).

For $N_{A,z}$, the flux in the z direction, we assume no changes in concentration $c_A(x,z)$ as a function of time (equilibrium), and neglect the contribution from diffusion of A in the z direction (small compared to the air velocity). In addition, diffusion at the edges of the pool in the y direction is neglected. Geometry of the surroundings may influence these assumptions (such as physical barriers above and to the side of the pool), but may be neglected if the concentration of the vapor is relatively low at that boundary. In order to simplify the analysis for the overall evaporation rate, the air velocity is also assumed constant as a function of x and flow only in the z direction. Thus,

$$N_{A,z} \approx C_A V_z \quad (2)$$

For $N_{A,x}$, the flux above the pool in the x direction, we neglect convective transport (no net flow in that direction). This essentially means that c_A is small, and that there is essentially no mixing in the area above the pool where the greatest portion of the concentration gradient exists. Thus,

$$N_{A,x} \approx -D_{A,B} \frac{\delta C_A}{\delta x} \quad (3)$$

where D_{AB} is the diffusion coefficient of chemical A in B (air). To be rigorous, the diffusion coefficient should be defined at the local temperature of the flowing air. However, this temperature will vary between the liquid surface temperature and the source air temperature. In this analysis, the liquid temperature is used.

Thus, equation (1) becomes:

$$V_z \frac{\delta C_A}{\delta z} = D_{AB} \frac{\delta^2 C_A}{\delta x^2} \quad (4)$$

The boundary conditions are as follows:

at $z = 0$	$C_A = 0$
$x = 0$	$C_A = C_{AO}$
$x = \infty$	$C_A = 0$

C_{AO} is assumed to be the concentration represented by the vapor pressure of the liquid at the liquid surface temperature. In reality, a temperature gradient will usually exist

between the liquid surface and the air temperature some distance away. Using the liquid vapor pressure neglects the effect of this gradient on the evaporation.

The solution to this partial differential equation is:

$$\begin{aligned} \frac{C_A}{C_{A0}} &= 1 - \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4D_{AB}z/V_z}}} e^{-\eta^2} d\eta \\ &= \operatorname{erfc}\left(\frac{x}{\sqrt{4D_{AB}z/V_z}}\right) \end{aligned} \quad (5)$$

The complementary error function (erfc) is well known as shorthand for the integral in equation (5) above.

This equation gives the vapor concentration as a function of z and x . The evaporation rate at any point on the liquid surface is given by:

$$N_{A,x}|_{x=0} = -D_{AB} \frac{\delta C_A}{\delta x} \quad (6)$$

The total evaporation rate is therefore the sum of all these points over the entire liquid surface, given by:

$$\begin{aligned} \text{Evap. Rate} &= 1/(\text{area of pool}) \iint N_{A,x} dz dy \\ &= 1/\text{area} \iint -D_{AB} \frac{\delta C_A}{\delta x} dz dy \end{aligned} \quad (7)$$

Solving the partial derivative in equation (7) by differentiating equation (5) yields:

$$\begin{aligned} N_{A,\text{total}} &= \frac{1}{\Delta y \Delta z} \iint \frac{2D_{AB}C_{A,0}}{\sqrt{4\pi D_{AB}z/V_z}} dz dy \\ &= 2C_{A,0} \sqrt{\frac{D_{AB}V_z}{\pi \Delta z}} \end{aligned} \quad (8)$$

This equation is in units of moles per unit time and area. In units of weight, we multiply by the molecular weight of the evaporating substance:

$$\text{Evap. Rate (mass/time + area)} = 2 \text{ MW } c_{A,0} \sqrt{\frac{D_{AB} V_z}{\pi \Delta z}} \quad (9)$$

The concentration of the evaporating solute at the interface is given by $c_{A,0}$, which can be approximated by:

$$c_{A,0} = \frac{\text{vapor pressure}}{R T} \quad (10)$$

where "vapor pressure" refers to the saturated vapor pressure above a liquid at temperature "T" in Kelvin, and "R" is the gas constant (39.381 in. Hg * ft³ / { lb moles * Kelvin }).

Equation (9) can therefore be written in the following dimensions:

$$\text{Evap. Rate (lb/hr ft}^2\text{)} = \frac{13.3792 \text{ M.W. V.P.}}{T} \sqrt{\frac{D_{AB} V_z}{\Delta z}} \quad (11)$$

where:

M.W. = molecular weight of evaporating liquid
 V.P. = vapor pressure (in. Hg)
 T = temperature (Kelvin)
 D_{AB} = diffusion coefficient of liquid vapor in air, ft^2/sec
 v_z = air velocity, ft/min
 Δz = pool length along flow direction, ft.

Equation (11) is completely general for a dilute evaporating substance in a flowing air stream.

The diffusion coefficient may not be available for many cases. According to Chapman-Enskog kinetic theory (Bird, Stewart and Lightfoot section 16.4), the diffusion coefficient is given by:

$$D_{AB}(\text{cm}^2/\text{sec}) = \frac{0.0018583 \sqrt{T^3(1/M_A + 1/M_B)}}{p \sigma_{AB}^2 \Omega_{AB}} \quad (12)$$

where:

T = temperature, Kelvin
 M_A = chemical A molecular weight
 M_B = chemical B (air) molecular weight (= 29)
 p = pressure (atmospheres)
 σ_{AB} = Lennard-Jones distance, angstroms
 Ω_{AB} = function of Lennard-Jones potential

The Lennard-Jones parameters are tabulated (appendix B, Bird, Stewart and Lightfoot) and can be roughly fit against temperature and molecular weight. This is shown in Figures 2, 3 and 4.

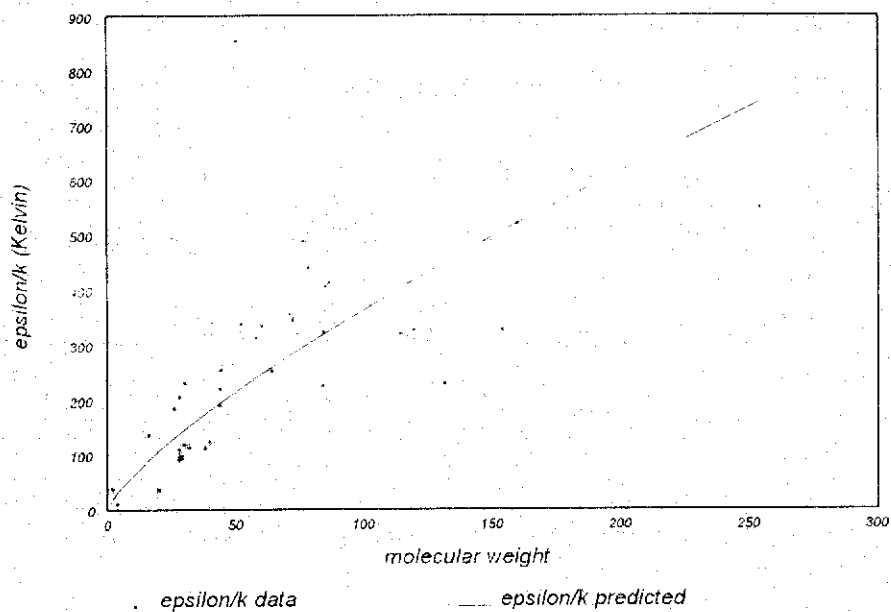


Figure 2

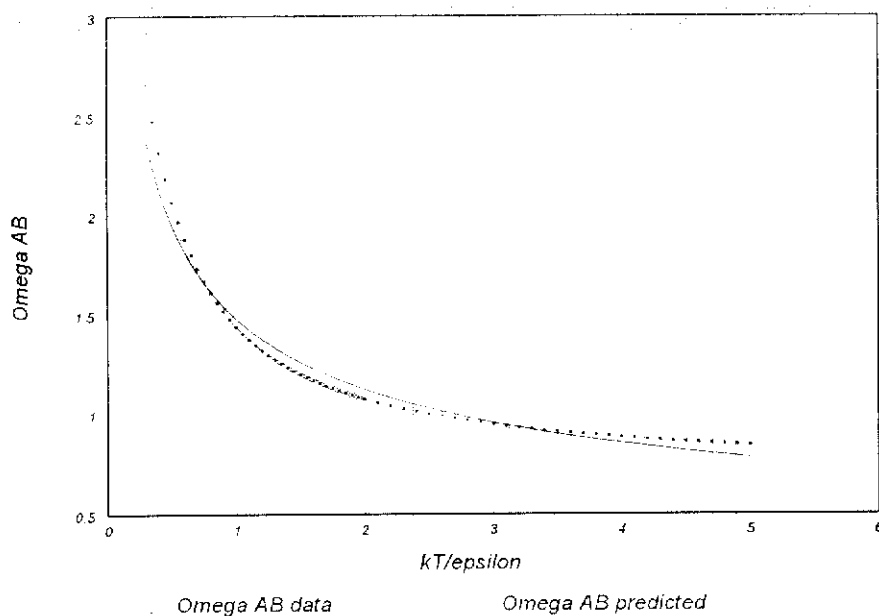


Figure 3

Figures 2 and 3 show the fit of the function Ω against molecular weight (Figure 2) and temperature (Figure 3). In Figure 2, the Lennard-Jones potential ϵ/k for a variety of compounds is shown versus their molecular weight. A simple exponential regression was fit against this graph with the results shown. Figure 3

shows the plot of the Omega function against the Lennard-Jones potential divided by temperature, and a simple exponential fit of this function. Combining these two fits gives an approximation of the Ω function in Equation 12.

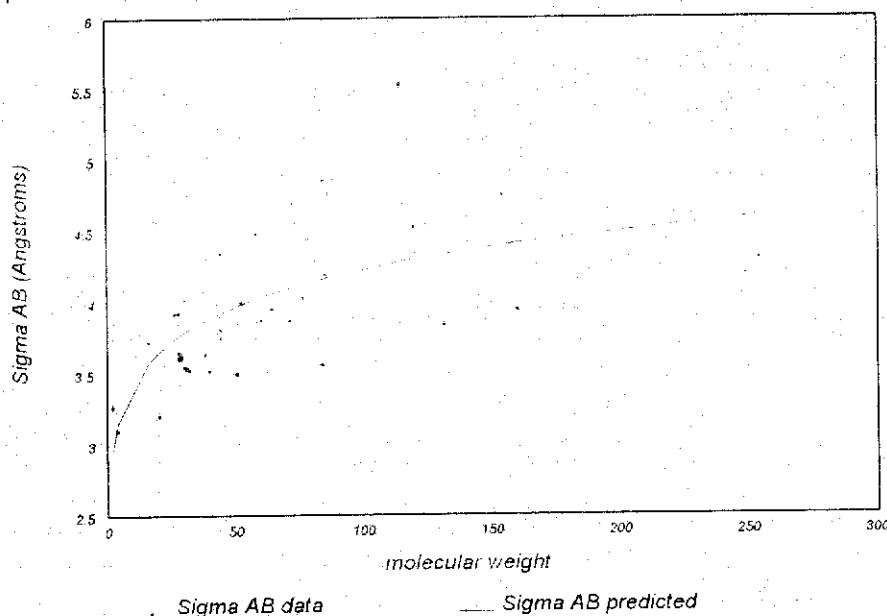


Figure 4

Similarly, Figure 4 shows the relationship and fit of data for the σ function in Equation 12. The exponential fits of these functions versus molecular weight and temperature are:

$$\begin{aligned}\sigma_{AB} &= e^{1.02} (M.W.)^{0.091} \\ \Omega &= e^{0.39} \left(\frac{\sqrt{97} e^{2.38} (M.W.)^{0.76}}{T} \right)^{0.397}\end{aligned}\quad (13)$$

Substituting into equation 12, this gives:

$$D_{AB}(\text{cm}^2/\text{sec}) = \frac{4.09 \times 10^{-5} T^{1.9} \sqrt{1/29 + 1/M.W._A} (M.W._A)^{0.33}}{p}\quad (14)$$

where:

p = pressure (atmospheres)

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T = temperature (Kelvin)
M.W._A = molecular weight of evaporating liquid

Converting D_{AB} to ft²/sec, and then inserting into equation (11), an approximate expression for the evaporation rate of a liquid in a flowing air stream is given by:

$$\text{Evap. Rate (lb/hr ft}^2\text{)} = \frac{2.79 \times 10^{-3} (M.W.)^{0.835} (V.P.)^4 \sqrt{1/M.W._A + 1/29}}{T^{0.05}} \sqrt{\frac{v_z}{\Delta z}} \quad (15)$$

where:

M.W. = molecular weight of evaporating liquid A
p = overall pressure, atmospheres
Δz = length of pool along air flow, feet
T = surface temperature of pool, Kelvin
V.P. = vapor pressure of substance A (in. Hg)
v_z = velocity of air, ft/min

Equation 15 is a completely general equation that can be used to predict the evaporation rate of a liquid in a flowing air stream. In addition, it takes into account the effect of changes in temperature, pressure, pool size and air velocity. Since it is rare to find completely stagnant air conditions in most spill situations, an approximation of the air speed, spill size can be made, and this equation will work well for estimating the evaporation rate. Although many assumptions were made to arrive at Equation 15, they are not at all unreasonable for a variety of situations. In general, the equation assumes low concentrations of the evaporating solvent, as compared to the air concentration in the room. As a result, this equation may begin to show errors for strongly evaporating solvents, long pool lengths or slow air wind velocities.

This equation can be tested by comparing it to experimental data provided by Pace Laboratories. In an extensive study using a specially-built apparatus, Pace measured the evaporation rate of 15 different compounds at several different temperatures and air velocities, and fit the data against "power law" regression against molecular weight, vapor pressure and air velocity, with generally good results. They performed an overall regression analysis for all chemicals except the "low vapor pressure" alcohols (1-hexanol, 1-heptanol and 2-octanol) and obtained the following equation:

$$\text{Evap. Rate (lb/hr ft}^2\text{)} = 0.000237 (MW) (VP) (V_z^{0.625}) \quad (16)$$

where:

MW = molecular weight of evaporating substance

VP = vapor pressure at liquid temperature, in. Hg
V_z = air velocity, ft/min

The EPA equation, based on work by Mackay is:

$$\text{Evap. Rate (lb/hr ft}^2\text{)} = \frac{0.03197(V_z^{0.78})(MW^{2/3})(VP)}{T} \quad (17)$$

In some cases, the wind velocity term has been assumed to be 10 mph, which corresponds to 880 feet/min. For the comparison below, however, the velocity term was included.

Figures 5, 6 and 7 compare Equations 15, 16 and 17 as applied to the Pace data, including the low boiling alcohols. The only differences between the figures are the scales, showing the experimentally measured evaporation rates from 0 to 0.06, 0 to 0.7 and 0 to 6 lbs/hr ft².

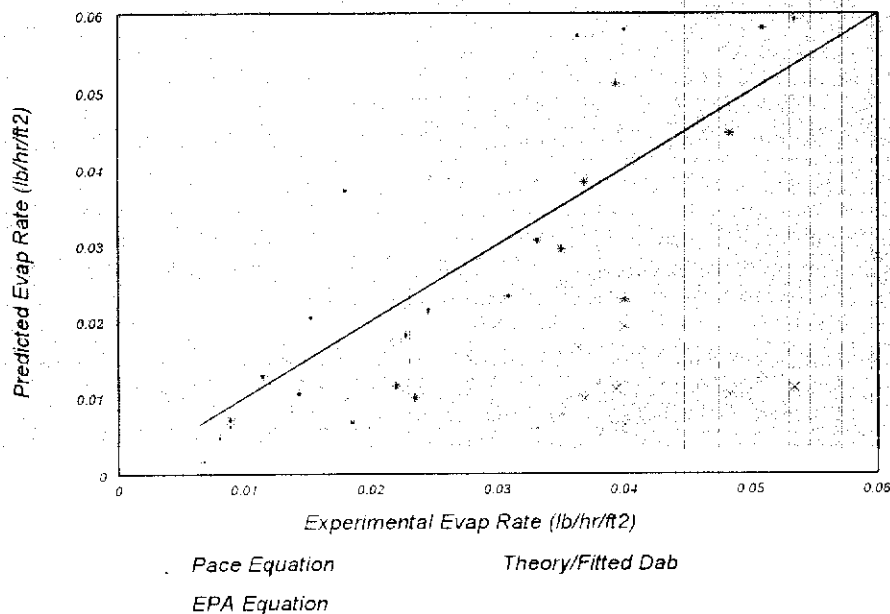


Figure 5

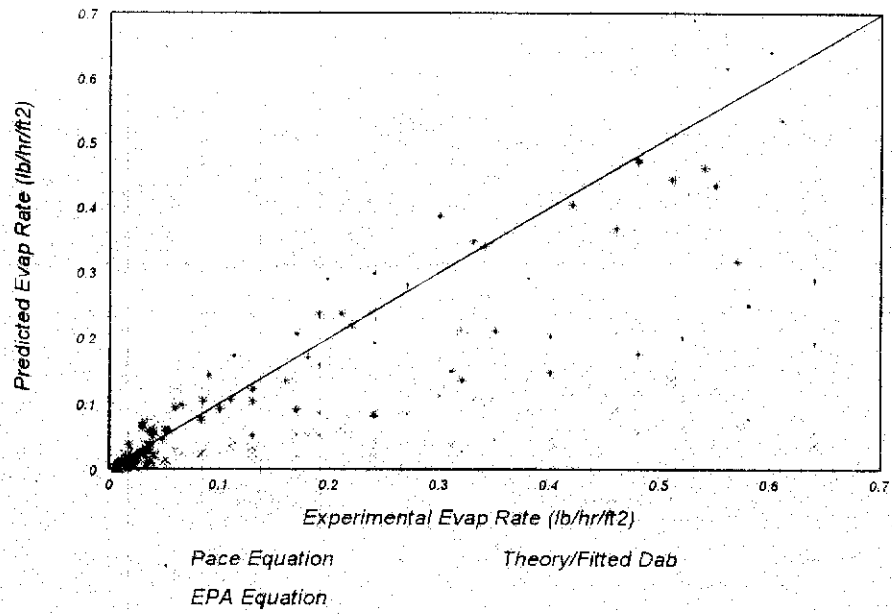


Figure 6

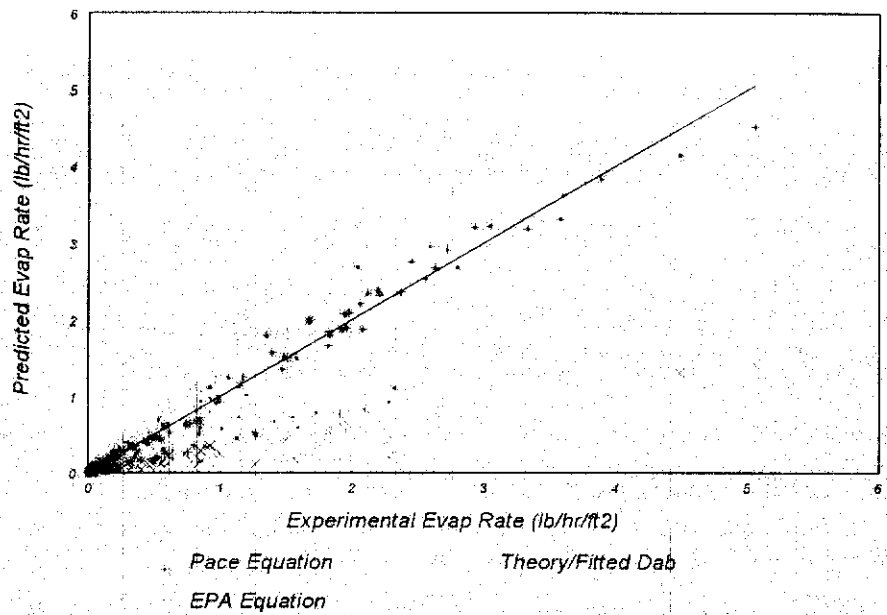


Figure 7

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A perfect fit of experiment and equation would be on the straight line along the diagonal. As can be seen in the three figures, both the Pace equation and the theoretical Equation 15 does a good job over a large range of evaporation rates. Since the Pace equation was a regression fit, it should naturally fit the closest. Remarkably, the theoretical equation is almost as good, and seems to be more accurate at the lower evaporation rates, which would correspond to normally encountered situations in EPA PMN analysis. The MacKay equation, however, is extremely poor, universally underestimating the actual evaporation rate by an order of magnitude for many measurements.

Closer inspection of the data shows that the theoretical equation 15 does the best job of the three equations at the low velocity gas rates of 100 feet/minute, but like the other two equations, it underestimates the experimentally measured evaporation rate for all of the low flow rate data. It is not clear whether this is from a deficiency in the equations or in the experiment. However, since these low rates would be more frequently encountered when judging evaporation rates inside ventilated buildings, the theoretical equation may be the best choice of the three. In addition, the assumptions are clear in the theoretical development, and cases where they may be violated can be easily identified. The agreement between the Pace results and the theory shows that these assumptions also apply to the Pace experiment.

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